A Network-Aware Predictive Digital Twin Framework for Event-Based Production Monitoring

Abstract. Discrete sensor events provide limited evidence for monitoring cyber-physical production systems, especially when observations are delayed, incomplete, or inconsistent with a predictive model. This paper presents a network-aware predictive monitoring framework that separates event prediction from evidence eligibility and anomaly interpretation. A process-specific predictor supplies an expected event-time window, while a reusable monitoring kernel evaluates timestamped binary transitions. Data freshness, edge-history continuity, and calibration applicability determine whether a model-based finding is admissible; unavailable evidence is reported separately from a process mismatch. Physics-based prediction, constant residual correction, and learned residual correction are interchangeable alternatives rather than assumed performance improvements. A conveyor-based MuJoCo cell serves as the single virtual testbed. A frozen Ridge model is assessed with separated training, validation, and calibration data. Its normal-test arrival-time mean absolute error is 15.4 milliseconds versus 18.1 milliseconds for constant correction, but its wider calibration window does not improve fault-warning timeliness. Across 31 recordings and 465 scheduled method replays, both corrected alternatives flag all 18 fault segments under immediate reception, with no findings in nine normal segments. Nine missing-sensor and stalled-drive pairs nevertheless have identical inputs, preventing unique fault attribution. The results support a reusable monitoring formulation and expose limits of learned prediction under incomplete observations. They do not establish transfer across machines, physical diagnostic accuracy, production recovery, or autonomous PLC reconfiguration.

Keywords: Predictive digital twin · Event-based monitoring · Data freshness · Incomplete observations · Residual learning

1 Introduction

Cyber-physical production systems execute operations whose progress is often represented by discrete events: a workpiece crosses a sensing point, an actuator reaches an endpoint, or an axis reports arrival. A monitoring decision must relate the received event to the operation that was expected and to the time at

which the observation was generated. Digital-twin research distinguishes model representation from actual physical–virtual integration [7]; production-system studies emphasize sensing, acquisition, and the services built on that information [8, 11, 2]. Jones et al. [6] further identify states, connections, and twinning rate among characteristics of a twin. These foundations motivate treating an observation's availability as part of a monitoring method rather than assuming that a simulated state is a measurement.

Communication delay and loss are established concerns in networked control [5], and age of information provides a time-based description of freshness [12]. Reviews discuss physical and learned models for digital twins [9, 4]. Neither a more detailed model nor a lower prediction error alone establishes that its output is suitable for a particular decision. Shao et al. [13] place verification, validation, and uncertainty assessment when assessing the credibility of a manufacturing twin. Lu et al. [14] examine how quality issues propagate through the development of digital twins. Their work supports evaluating the data and model basis of a service; it does not validate the thresholds used in the present testbed.

Prior applications use twins to generate training data for manufacturing fault diagnosis [1] and to assess how manufacturing lines can be reconfigured under disturbances [3]. The present paper asks a narrower question: *when can a predicted event window support an interpretable finding of inconsistent process progress under incomplete observations, and does a learned residual improve that finding over a simple correction?* Prediction error, warning timeliness, evidence availability, and fault attribution are evaluated as distinct outcomes. A missing transition is not automatically assigned to a uniquely identified physical fault.

We propose a process-independent event-window monitoring interface with explicit eligibility conditions, and implement a reusable binary-event kernel behind a process-specific prediction adapter. A matched virtual comparison evaluates a pulse-based predictor, a training-mean correction, and Ridge residual learning. The cell is a testbed for this formulation, not the definition of the method. The intended abstraction includes other predictable binary production events, but the empirical evaluation concerns one conveyor-based system only. This is an exploratory methodological case study, not a systematic literature review, a claim of general algorithmic novelty, or a demonstration of autonomous production recovery.

2 Problem Formulation and Applicability

Consider a production operation with a past, observable anchor at time t_a and a subsequent binary completion event e. A predictor P_m uses admissible observations and issued commands up to t_a to produce an event-time estimate $\widehat{t}_m$ and a window $I_m = [\ell_m, u_m]$. The anchor need not be a conveyor pulse: it can be an observed operation start or another validated transition. The predictor and its calibration are specific to that event and operating domain.

An observation is represented as

$$o_k = (s_k, r_k, n_k, y_k, q_k), \qquad A_k = t_k - s_k, \tag{1}$$

where s_k is source time, r_k reception time, n_k sequence number, y_k the binary event value, q_k quality metadata, and t_k decision time. Define eligibility as the conjunction

$$G_k = G_{\text{clock}} \wedge G_{\text{age}} \wedge G_{\text{quality}} \wedge G_{\text{history}} \wedge G_{\text{domain}}. \tag{2}$$

These terms require compatible clocks with $s_k \leq r_k \leq t_k$, age within the configured limit, resolved signal quality, sufficient contiguous edge history, and applicability of the prediction and calibration. Missing metadata cannot silently be replaced by an assumed healthy value. Here, clock compatibility is provided by the shared simulation clock; physical clock alignment is not implemented.

The formulation requires a predictable event, a causal anchor, a calibrated operating domain, and an explicit relation between local commands and the observed operation. It is unsuitable when the expected event cannot be predicted from available information, when unknown command changes invalidate the forecast, or when competing operations cannot be distinguished. One monitor instance covers one operation; overlapping products and automatic reassignment of event identity are outside the evaluated implementation.

3 Monitoring Method

3.1 Separation of Prediction and Event Evaluation

The predictor supplies a window; the event kernel consumes that window and eligible binary evidence. The kernel contains no conveyor coordinates, actuator equations, robot geometry, PLC addresses, or MuJoCo calls. Such information belongs in the process adapter. Consequently, changing equipment does not require changing the event comparison logic, but it does require establishing a new observation mapping, predictor, calibration, and eligibility assumptions. Interface reuse is not evidence that an existing calibrated model transfers between machines.

For a point predictor and an empirical residual radius, the implemented window is

$$I_m = [\max(t_a, \widehat{t}_m - r_m), \widehat{t}_m + r_m], \quad r_m = \max_{i \in \mathcal{D}_{\text{cal}}} |t_{e,i} - \widehat{t}_{m,i}|. \tag{3}$$

The radius uses a separate normal calibration set. It is an observed maximum on that set, not a confidence bound or a calibrated probability of a future event. This choice is deliberately simple so that differences in predictor error and window width remain inspectable.

3.2 Findings, Abstention, and Evidence Attribution

For a resolved rising edge at source time s_k, a finding is early if $s_k < \ell_m$, late if $s_k > u_m$, and otherwise consistent with the window. If no target edge has been seen, missing progress is flagged only after the configured number of consecutive

eligible samples satisfy $s_k > u_m$. The case study uses two samples. A high value at the first eligible observation cannot retrospectively establish the time of its rising edge and is labeled unresolved.

When G_k is false, the monitor reports insufficient evidence rather than a healthy process or an inferred missing event. Lost edge history invalidates the current kernel instance; later reception alone does not restore its missing history. The adapter can select a separate engineering-bound predictor when an empirical model is outside its domain, but that result retains a different attribution. In particular, an engineering fallback alert must not be credited to a learned predictor.

The source-time condition matters under delay: lack of a received edge at decision time does not establish absence through that time. An old packet only reports the process through its source timestamp. The monitored outcome is therefore an inconsistency of expected progress with admissible evidence, not necessarily a classified sensor, communication, or drive fault. No monitor result has actuator or PLC output authority in this study.

3.3 Causal Learning and a Matched Simple Reference

A learned residual can supplement a nominal event-time estimate:

$$\widehat{t}_{\text{learned}} = \widehat{t}_{\text{base}} + b + w^T \widetilde{\phi}. \tag{4}$$

Features ϕ must be available before the target event; scaling, fitting, model selection, and envelope calibration use their assigned data partitions. The learned alternative is compared with the same baseline plus a training-mean residual. This reference tests whether apparent learning benefit is principally a systematic offset correction. A learned model is not selected merely because its point-prediction error is lower: the resulting window, false findings, rejection rate, and warning time also matter.

4 Virtual Testbed and Experimental Protocol

4.1 Process-Specific Adapter

A conveyor-based production cell supplies a single virtual case study. MuJoCo provides the multibody/contact simulation environment [10]; the monitoring inputs are separated from simulator pose and hidden fault labels. The broader cell includes a pusher and gantry, but the reported experiment isolates one workpiece moving between two optical sensing locations. It does not evaluate the complete pick-and-place cycle. CAD V5 and the functional MuJoCo baseline remain different geometries and are not presented as synchronized representations of a calibrated physical machine.

The adapter uses the completed upstream occupancy pulse as its anchor. If L is nominal workpiece length, D_p commanded travel integrated during the pulse, v constant issued speed, and d the remaining event-to-event travel, then

$$\widehat{g} = L/D_p, \qquad \widehat{t}_{\text{pulse}} = t_a + d/(v\widehat{g}). \tag{5}$$

The Ridge features are pulse duration, issued speed, and their product, frozen at pulse completion. Labels are sampled downstream optical-edge timestamps, not simulator positions. Only constant forward command segments are admitted to this empirical comparison. Source sampling is 20 ms; preliminary workpiece length is 60 mm. Neither number constitutes a new physical calibration campaign.

The adapter checks the training feature box and the stored calibration context. Changes in workpiece length, sensor spacing, sampling period, or command history prevent reuse of the old empirical window. An engineering pulse-bound fallback remains separate; invalid quality or lost history suppresses model findings. These checks do not establish complete domain validity or a physical risk guarantee.

4.2 Data Partitions and Frozen Comparisons

The deterministic normal campaign contains 61 segments: 32 training, eight validation, eight error-envelope calibration, nine normal test, and four shifted-condition segments. Speed–gain–friction tuples do not overlap between partitions. However, the admitted observation–command pairs are distinct only 16, four, four, nine, and four times, respectively, because the friction variants in the first three partitions produce identical recordings. No pair overlaps between partitions. These designed conditions and duplicates are not independent factory observations.

Ridge penalties 0.01, 0.1, 1, and 10 are selected by validation mean absolute error (MAE); scaling and weights remain fitted on training rows. The model is sealed before calibration and test generation. A constant-bias reference was added after the initial pilot and is disclosed as post hoc; its offset uses training rows only. Ridge features, penalty selection, and weights were not revised from test outcomes. A first generation attempt failed before a usable training segment because the fixture inherited an incompatible degraded-speed setting; the local fixture correction and failed attempt remain documented.

Monitoring evaluation retrospectively reuses the nine normal test and four shifted recordings and adds missing-target feedback and a stalled drive for each normal test condition: 31 recordings in total. Three methods and five scheduled delivery cases produce 465 replays. Cases are immediate reception, 40 ms delay, 120 ms stale delivery, a brief post-pulse gap, and unavailable target-signal quality. Model weights and separately calibrated radii remain fixed. This is exploratory replay, not a new independent confirmation set or a measurement of physical network performance.

Outcomes are point-prediction error, findings in normal segments, findings in faulty segments, evidence rejection, fallback attribution, and paired warning time. A recording ending before its required confirmation window is censored rather than silently classified as healthy or a proven missed detection. Fault injection time and simulator state are evaluator-only information. Replays of the same recording are matched software conditions, not extra independent trials.

Table 1. Sampled optical-edge arrival-time prediction: held-out normal test ($n = 9$) and shifted conditions ($n = 4$). MAE in ms; not physical timing accuracy.

Method	Test MAE	Shift MAE
Nominal command kinematics	105.7	330.0
Upstream pulse kinematics	98.5	83.3
Pulse plus train-mean bias	18.1	28.5
Pulse plus Ridge residual	15.4	38.3

Constant-bias reference added after the initial pilot; its value uses training rows only.

Table 2. Advisory ETA findings in fixed replay conditions; fault denominator combines nine missing-target and nine stalled-drive segments, which form nine identical input pairs.

Method	Radius (ms)	Normal alerts no delay (/9)	Fault alerts no delay (/18)	Fault alerts 40 ms (/18)
Pulse	140.0	0	18	16
Constant bias	42.9	0	18	18
Ridge	65.0	0	18	18

Two delayed pulse replays end before a late-edge confirmation window. Data rejection and engineering fallback are not credited as learned-model detections.

5 Results

5.1 Prediction Error Is Not Warning Timeliness

On the nine normal test segments, Ridge MAE is 15.4 ms versus 18.1 ms for constant correction and 98.5 ms for the uncorrected pulse estimate. Its incremental advantage over the simple corrected reference is only about 2.7 ms, without a significance claim. Under the four shifted conditions, Ridge is worse than constant correction (38.3 versus 28.5 ms); all four lie outside the training feature box. Sampling and the small designed grid preclude inferring physical sub-sample timing accuracy.

The calibration radii are 140.0 ms for the pulse estimate, 42.9 ms for constant correction, and 65.0 ms for Ridge. Thus the learned point estimate has lower normal-test MAE but a wider event window than the corrected reference. The eight calibration rows contain only four distinct admitted recording pairs, and these radii are not statistical coverage guarantees.

5.2 Findings and Unavailable Evidence

Under immediate reception, each method flags all 18 fault segments and none of the nine normal segments. Under 40 ms delay, both corrected alternatives flag 18 of 18; the uncorrected pulse method flags 16. The two remaining pulse replays end before a late-edge confirmation window and therefore lack sufficient

observation for a missed-detection conclusion. Under usable delivery, the four shifted normal recordings use engineering fallback and produce no findings; these are not learned-model successes.

Relative to constant correction, Ridge warns at the same time in two immediate fault replays and 20–40 ms later in 16; it is never earlier. The 40 ms delivery case preserves that ordering. This descriptive result illustrates why improvement in a point-prediction metric cannot be substituted for improvement in a decision metric.

All stale-delivery inputs are rejected before a usable fit. Post-pulse gaps and target-quality loss stop admissible evaluation after initial prediction. Zero model findings in these cases reflects insufficient evidence, not correct recognition of healthy operation. All nine missing-sensor/stalled-drive pairs have identical admitted observation and issued-command hashes. The findings consequently cannot uniquely attribute the physical cause, and the 18 fault segments do not represent 18 independent observation patterns.

6 Generality, Limitations, and Focused Development

The general element is the separation of a causal event predictor, evidence eligibility, and event-window evaluation. A cylinder-endpoint adapter could anchor at an observed stroke start; an axis-arrival adapter could use a validated motion start and commanded trajectory. Each would need its own event semantics, feedback mapping, dynamic assumptions, and calibration. Their mention describes interface applicability, not validation on those machines. Software contract checks with renamed event labels do not constitute additional physical or simulated-device experiments.

The packaged runtime selects the monitoring method and its MuJoCo case-study dependencies. Full-cell reconfiguration, CAD authoring, graphics, reinforcement learning, and whole-cell online contact forecasting are outside the active paper scope. Historical code, failed and adverse results, parameterized profiles, and the last user-accepted image remain preserved. Scope reduction does not convert earlier timing or recovery limitations into successful results.

The main evidence limitations are one virtual process, deterministic and partly duplicated conditions, reuse of previously inspected normal test recordings, a post hoc simple reference, assumed rather than physically identified dynamics, and empirical calibration maxima. The study establishes neither cross-machine transfer nor live clock synchronization, diagnostic accuracy, hard real-time execution, production availability, or successful operation after loss of required feedback. Model/source hashes and reproducibility checks support traceability, not scientific novelty or a guarantee of acceptance by a venue.

The next evaluation should freeze a fresh condition set and compare source-aware monitoring with an explicitly defined timestamp-naive reference under matched information and a stated false-finding budget. It should also evaluate a second event adapter and estimate physical model/measurement uncertainty

when measurements become available. Until those steps are completed, the general interface and single-testbed results must remain separate claims.

7 Physical Validation

Reserved: no physical measurements or validation results are reported.

7.1 Physical Setup and Calibration

7.2 Live Synchronization and Network Measurements

7.3 Physical Event-Monitoring Trials

8 Conclusion

A network-aware predictive monitoring framework has been formulated around binary production events and implemented with a process-independent comparison kernel. One conveyor-based MuJoCo testbed demonstrates the distinction between a model mismatch, unavailable evidence, and ambiguous fault attribution. Learned residual correction improves normal-test event-time MAE slightly over a constant correction but does not improve warning timeliness and performs worse under shifted conditions. The supported outcome is an inspectable event-monitoring method and a bounded exploratory comparison, not verified cross-machine resilience or autonomous PLC reconfiguration.

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References

1. Ademujimi, T., Prabhu, V.: Digital twin for training Bayesian networks for fault diagnostics of manufacturing systems. Sensors **22**(4), 1430 (2022). https://doi.org/10.3390/s22041430
2. Cimino, C., Negri, E., Fumagalli, L.: Review of digital twin applications in manufacturing. Computers in Industry **113**, 103130 (2019). https://doi.org/10.1016/j.compind.2019.103130

3. Fu, B., Bi, M., Umeda, S., et al.: Digital twin-based smart manufacturing: Dynamic line reconfiguration for disturbance handling. IEEE Transactions on Automation Science and Engineering **22**, 14892–14905 (2025). https://doi.org/10.1109/TASE.2025.3563320
4. Fuller, A., Fan, Z., Day, C., Barlow, C.: Digital Twin: Enabling Technologies, Challenges and Open Research. IEEE Access **8**, 108952–108971 (2020). https://doi.org/10.1109/ACCESS.2020.2998358
5. Hespanha, J.P., Naghshtabrizi, P., Xu, Y.: A survey of recent results in networked control systems. Proceedings of the IEEE **95**(1), 138–162 (2007). https://doi.org/10.1109/JPROC.2006.887288
6. Jones, D., Snider, C., Nassehi, A., Yon, J., Hicks, B.: Characterising the Digital Twin: A systematic literature review. CIRP Journal of Manufacturing Science and Technology **29**, 36–52 (2020). https://doi.org/10.1016/j.cirpj.2020.02.002
7. Kritzinger, W., Karner, M., Traar, G., Henjes, J., Sihn, W.: Digital Twin in manufacturing: A categorical literature review and classification. IFAC-PapersOnLine **51**(11), 1016–1022 (2018). https://doi.org/10.1016/j.ifacol.2018.08.474
8. Negri, E., Fumagalli, L., Macchi, M.: A Review of the Roles of Digital Twin in CPS-based Production Systems. Procedia Manufacturing **11**, 939–948 (2017). https://doi.org/10.1016/j.promfg.2017.07.198
9. Rasheed, A., San, O., Kvamsdal, T.: Digital Twin: Values, Challenges and Enablers From a Modeling Perspective. IEEE Access **8**, 21980–22012 (2020). https://doi.org/10.1109/ACCESS.2020.2970143
10. Todorov, E., Erez, T., Tassa, Y.: MuJoCo: A physics engine for model-based control. In: 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 5026–5033. IEEE (2012). https://doi.org/10.1109/IROS.2012.6386109
11. Uhlemann, T.H.-J., Lehmann, C., Steinhilper, R.: The Digital Twin: Realizing the Cyber-Physical Production System for Industry 4.0. Procedia CIRP **61**, 335–340 (2017). https://doi.org/10.1016/j.procir.2016.11.152
12. Yates, R.D., Sun, Y., Brown, D.R., III, Kaul, S.K., Modiano, E., Ulukus, S.: Age of information: An introduction and survey. IEEE Journal on Selected Areas in Communications **39**(5), 1183–1210 (2021). https://doi.org/10.1109/JSAC.2021.3065072
13. Shao, G., Hightower, J., Schindel, W.: Credibility consideration for digital twins in manufacturing. Manufacturing Letters **35**, 24–28 (2023). https://doi.org/10.1016/j.mfglet.2022.11.009
14. Lu, Y., Yang, Z., Li, S., Zhao, Y.F., Xie, J., Safdar, M., Ko, H.: An overarching quality evaluation framework for additive manufacturing digital twin. In: 2024 IEEE 20th International Conference on Automation Science and Engineering (CASE). IEEE (2024). https://doi.org/10.1109/CASE59546.2024.10711607